

Designing and Securing a Small Business Network Infrastructure

Executive Summary

This project focused on improving the security of a small business network that lacked proper segmentation and access control. Initially, all devices were on a single network, allowing unrestricted communication and increasing the risk of unauthorized access and data breaches.

To address these issues, a secure network architecture was designed and implemented using VLAN segmentation, inter-VLAN routing, and access control lists (ACLs). Cisco Packet Tracer was used to simulate the network environment and apply configurations.

Testing results showed that communication between networks was successfully controlled, and unauthorized access from the guest network to the admin network was blocked. The project demonstrates the effectiveness of basic network security controls in protecting organizational infrastructure.

Project Planning & Requirements

Project Overview

The project aimed to design and implement a secure network for a small business environment by identifying vulnerabilities and applying appropriate cybersecurity controls.

Timeline

Phase	Description
Week 1	Project proposal and planning
Week 2	Network design and VLAN setup
Week 3	Implementation and configuration
Week 4	Testing, documentation, and reporting

Milestones

- Proposal completed
- Network design finalized
- VLAN and routing configured
- ACL implemented
- Testing completed
- Final report completed

Technical Requirements

- Cisco Packet Tracer
- Router and Switch devices
- End-user PCs
- CLI configuration tools

Functional Requirements

- Segment network into multiple VLANs
- Enable communication between VLANs
- Restrict unauthorized access
- Test connectivity and security

Non-Functional Requirements

- Security
- Reliability
- Scalability
- Performance

Methodology & Solution Design

Problem Statement

The small business network lacked segmentation and access control, allowing all devices to communicate freely. This exposed sensitive systems to unauthorized users, including guest devices.

High-Level Design

The network was designed with:

- One router for inter-VLAN routing
- One switch for VLAN segmentation
- Four PCs representing different departments

VLANs created:

- VLAN 10 – Admin
- VLAN 20 – HR
- VLAN 30 – IT
- VLAN 40 – Guest

Low-Level Design

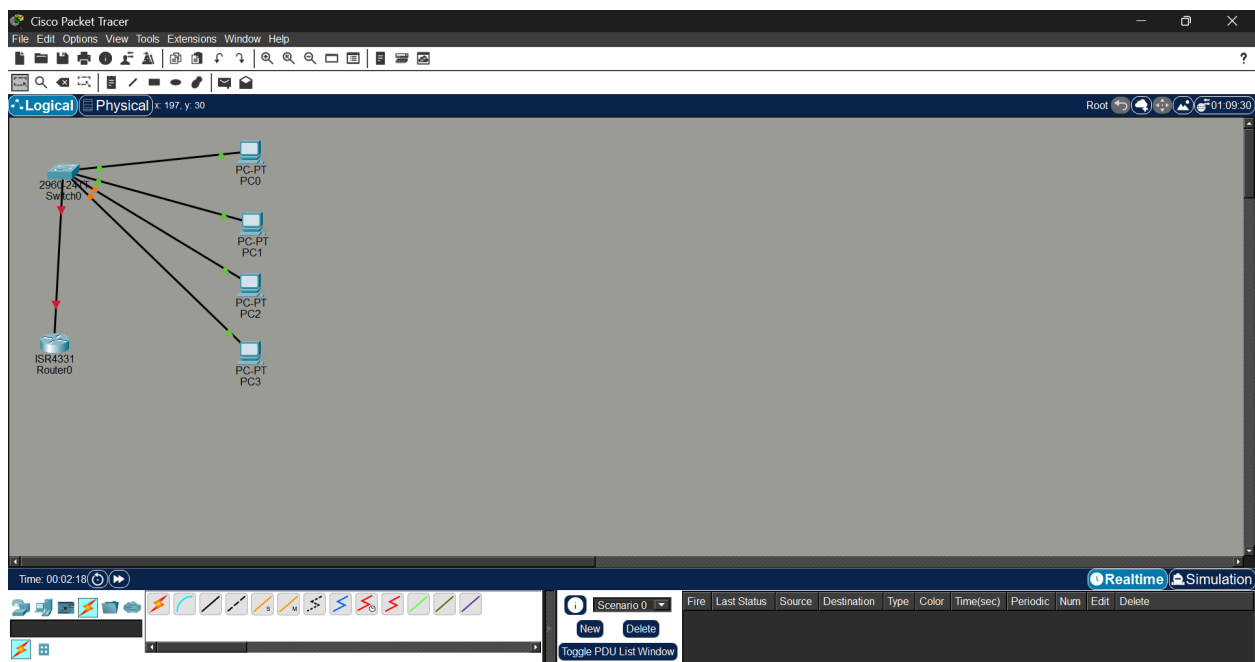
- Router-on-a-stick configuration using subinterfaces
- IP addressing scheme:
 - 192.168.10.0/24 (Admin)
 - 192.168.20.0/24 (HR)
 - 192.168.30.0/24 (IT)
 - 192.168.40.0/24 (Guest)
- Switch configured with:
 - VLAN assignments per port
 - Trunk link to router
- ACL configured to restrict Guest access to Admin network

Implementation

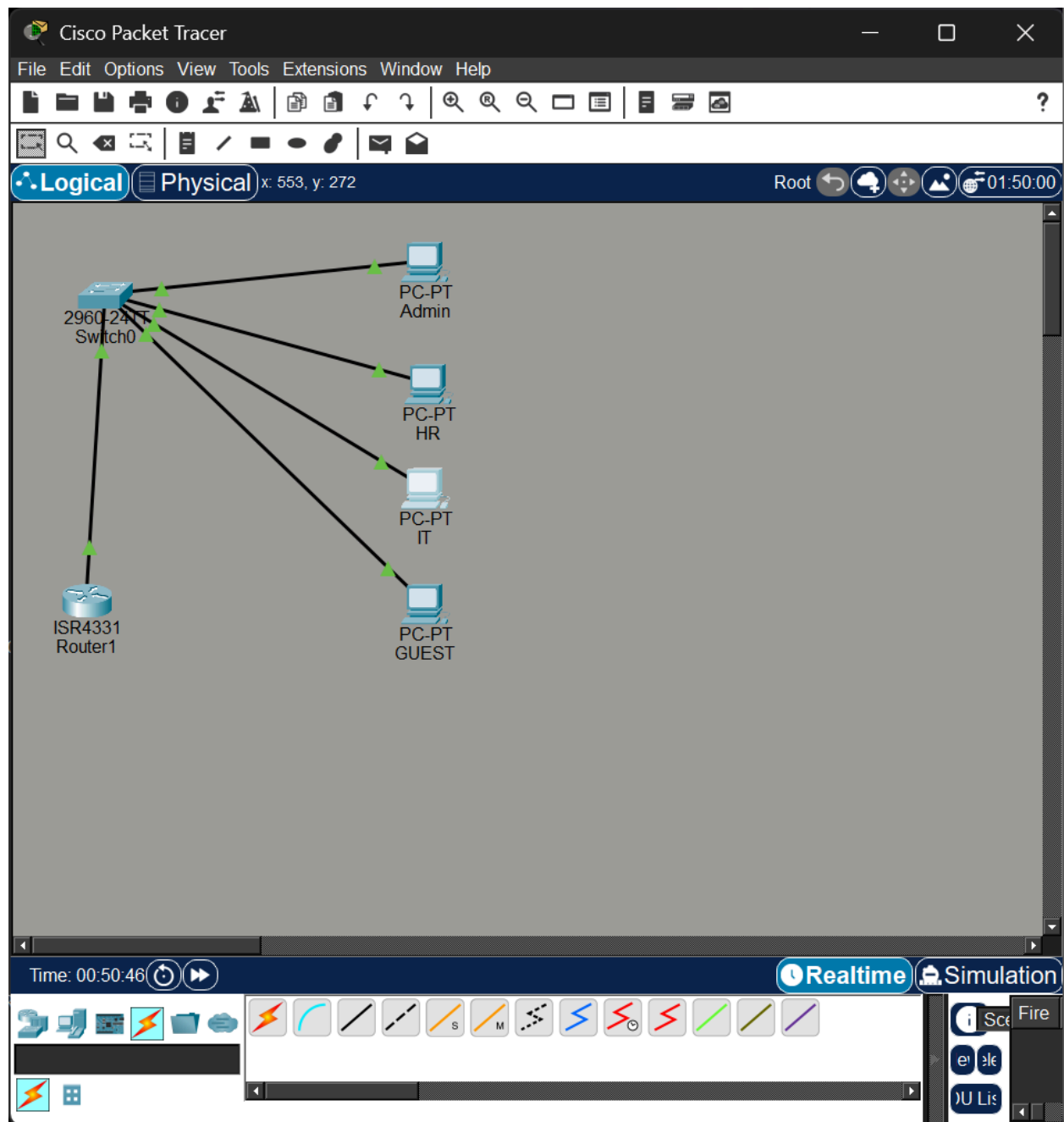
This section describes the step-by-step process used to configure and secure the network environment using Cisco Packet Tracer.

Network Setup

Devices were added and connected using copper straight-through cables. A router, switch, and four PCs were used.



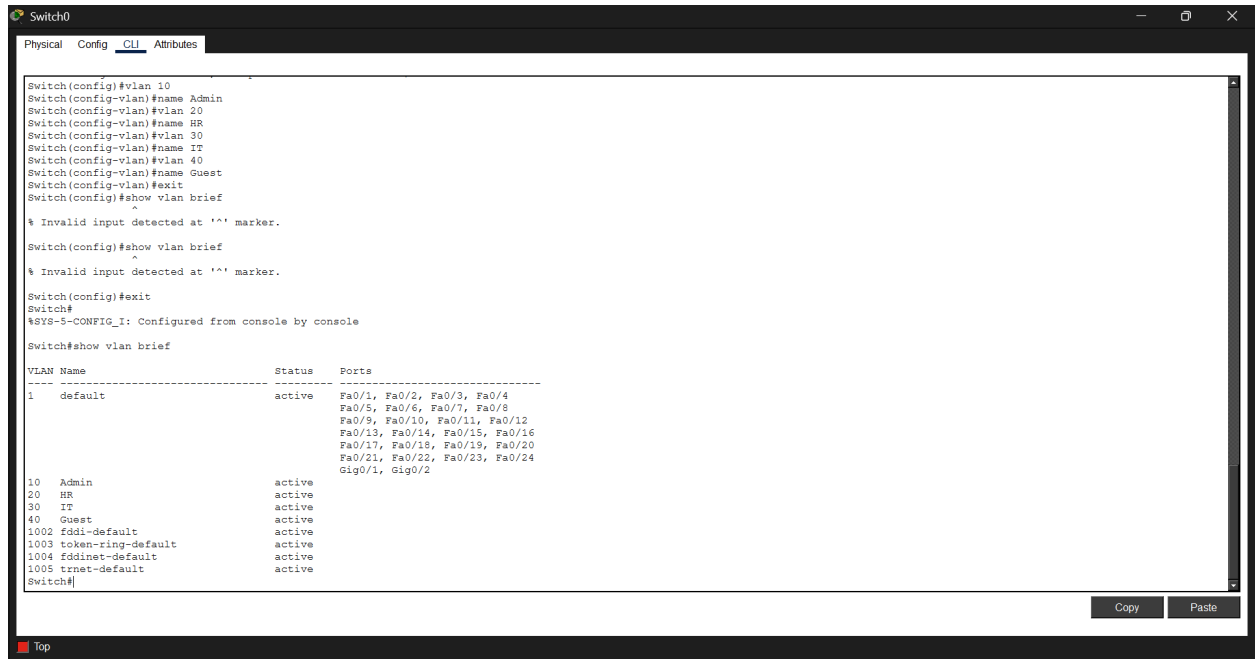
Initial Network Topology



Final Topology

VLAN Configuration

VLANs were created on the switch and assigned to ports.



```
Switch0
Physical Config CLI Attributes
Switch(config)#vlan 10
Switch(config-vlan)#name Admin
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name HR
Switch(config-vlan)#vlan 30
Switch(config-vlan)#name IT
Switch(config-vlan)#vlan 40
Switch(config-vlan)#name Guest
Switch(config-vlan)#exit
Switch(config)#show vlan brief

% Invalid input detected at '^' marker.

Switch(config)#show vlan brief

% Invalid input detected at '^' marker.

Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                           Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                           Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                           Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                           Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                           Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                           Gig0/1, Gig0/2
10   Admin                  active
20   HR                    active
30   IT                    active
40   Guest                 active
1002 fddi-default          active
1003 token-ring-default    active
1004 fddinet-default       active
1005 trnet-default         active
Switch#
```

VLAN Configuration (show vlan brief)

Trunk Configuration

A trunk link was configured between the switch and router.

The screenshot shows a network switch CLI window titled "Switch0". It has tabs for "Physical", "Config", "CLI", and "Attributes", with "CLI" selected. The terminal output shows the following sequence of commands and responses:

```
Switch#show interfaces trunk

Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#show interfaces trunk
      ^
% Invalid input detected at '^' marker.

Switch(config)#interface fa0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#show interfaces trunk
      ^
% Invalid input detected at '^' marker.

Switch(config-if)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show interfaces trunk

Switch#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Switch#show interfaces trunk
Port      Mode      Encapsulation  Status        Native vlan
Fa0/1     on        802.1q         trunking      1

Port      Vlans allowed on trunk
Fa0/1     1-1005

Port      Vlans allowed and active in management domain
Fa0/1     1,10,20,30,40

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     1,10,20,30,40

Switch#
```

At the bottom right of the terminal window are "Copy" and "Paste" buttons. At the bottom left of the window is a "Top" button with a red square icon.

Trunk Configuration (show interfaces trunk)

Router Configuration

Subinterfaces were created on the router to enable inter-VLAN routing.

Router1

Physical Config CLI Attributes

```
Router(config-subif)#ip address 192.168.30.1 255.255.255.0
Router(config-subif)#
Router(config-subif)#interface g0/0/0.40
Router(config-subif)#encapsulation dot1Q 40
Router(config-subif)#ip address 192.168.40.1 255.255.255.0
Router(config-subif)#
Router(config-subif)#end
%LINK-3-UPDOWN: Interface GigabitEthernet0/0/0.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0.10, changed state
to up

%LINK-3-UPDOWN: Interface GigabitEthernet0/0/0.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0.20, changed state
to up

%LINK-3-UPDOWN: Interface GigabitEthernet0/0/0.30, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0.30, changed state
to up

%LINK-3-UPDOWN: Interface GigabitEthernet0/0/0.40, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0.40, changed state
to up

Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip interface brief
Interface                IP-Address      OK? Method Status          Protocol
GigabitEthernet0/0/0     unassigned      YES unset        up              up
GigabitEthernet0/0/0.10  192.168.10.1    YES manual      up              up
GigabitEthernet0/0/0.20  192.168.20.1    YES manual      up              up
GigabitEthernet0/0/0.30  192.168.30.1    YES manual      up              up
GigabitEthernet0/0/0.40  192.168.40.1    YES manual      up              up
GigabitEthernet0/0/1     unassigned      YES unset        administratively down down
GigabitEthernet0/0/2     unassigned      YES unset        administratively down down
Vlan1                    unassigned      YES unset        administratively down down
Router#
```

Copy Paste

Top

Router Subinterface Configuration

IP Addressing

Each PC was assigned an IP address and default gateway corresponding to its VLAN.

The screenshot shows a configuration window titled "Admin" with tabs for Physical, Config, Desktop, Programming, and Attributes. The "Desktop" tab is active, and the "IP Configuration" section is highlighted. The interface is for "FastEthernet0".

IP Configuration

Interface: FastEthernet0

IP Configuration:

- ☒ DHCP
- ☒ Static

IPv4 Address: 192.168.10.10

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.10.1

DNS Server: 0.0.0.0

IPv6 Configuration

- ☒ Automatic
- ☒ Static

IPv6 Address: /

Link Local Address: FE80::260:5CFF:FE52:4AD4

Default Gateway:

DNS Server:

802.1X

☐ Use 802.1X Security

Authentication: MD5

Username:

Password:

☐ Top

Admin IP configuration

HR

Physical

Config

Desktop

Programming

Attributes

IP Configuration

InterfaceFastEthernet0

IP Configuration

DHCP

Static

IPv4 Address192.168.20.10

Subnet Mask255.255.255.0

Default Gateway192.168.20.10

DNS Server0.0.0.0

IPv6 Configuration

Automatic

Static

IPv6 Address/

Link Local AddressFE80::200:CFF:FED4:202B

Default Gateway

DNS Server

802.1X

Use 802.1X Security

AuthenticationMD5

Username

Password

Top

HR IP configuration

IT

PhysicalConfigDesktopProgrammingAttributes

IP Configuration

InterfaceFastEthernet0

IP Configuration

DHCP

Static

IPv4 Address192.168.30.10

Subnet Mask255.255.255.0

Default Gateway192.168.30.10

DNS Server0.0.0.0

IPv6 Configuration

Automatic

Static

IPv6 Address/

Link Local AddressFE80::201:64FF:FE8B:8504

Default Gateway

DNS Server

802.1X

Use 802.1X Security

AuthenticationMD5

Username

Password

Top

IT IP configuration

GUEST

Physical

Config

Desktop

Programming

Attributes

IP Configuration

X

Interface

FastEthernet0

IP Configuration

DHCP

Static

IPv4 Address

192.168.40.10

Subnet Mask

255.255.255.0

Default Gateway

192.168.40.10

DNS Server

0.0.0.0

IPv6 Configuration

Automatic

Static

IPv6 Address

/

Link Local Address

FE80::202:16FF:FE71:59

Default Gateway

DNS Server

802.1X

Use 802.1X Security

Authentication

MD5

Username

Password

Top

Guest IP configuration

Connectivity Testing (Before Security)

Initial testing confirmed that all VLANs could communicate with each other.

```
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time=16ms TTL=255
Reply from 192.168.10.1: bytes=32 time=1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 16ms, Average = 4ms

C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.10: bytes=32 time=11ms TTL=127
Reply from 192.168.20.10: bytes=32 time=1ms TTL=127
Reply from 192.168.20.10: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 11ms, Average = 4ms

C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time=11ms TTL=127
Reply from 192.168.20.10: bytes=32 time=11ms TTL=127

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 5ms

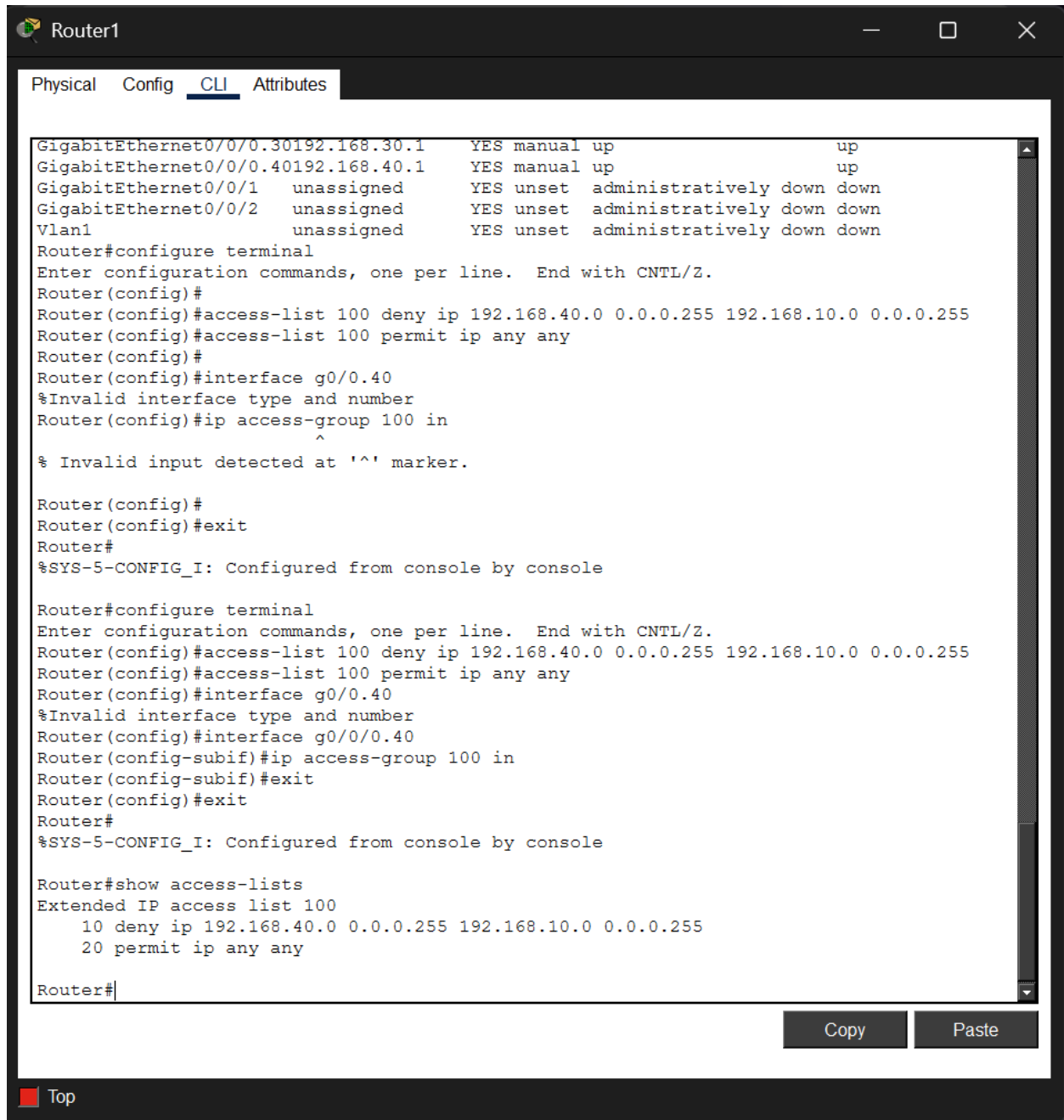
C:\>
```

Ping Test Before Security (Successful Communication)

This test confirms that legitimate traffic between authorized VLANs is still allowed after applying the ACL. This ensures that security controls do not disrupt normal network operations.

ACL Implementation

An ACL was configured to block Guest network access to the Admin network.



The screenshot shows a Cisco Router CLI window titled "Router1". The window has tabs for "Physical", "Config", "CLI", and "Attributes", with "CLI" currently selected. The CLI output shows the status of various interfaces and the configuration of an ACL.

```
GigabitEthernet0/0/0.30 192.168.30.1 YES manual up up
GigabitEthernet0/0/0.40 192.168.40.1 YES manual up up
GigabitEthernet0/0/1 unassigned YES unset administratively down down
GigabitEthernet0/0/2 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#access-list 100 deny ip 192.168.40.0 0.0.0.255 192.168.10.0 0.0.0.255
Router(config)#access-list 100 permit ip any any
Router(config)#
Router(config)#interface g0/0.40
%Invalid interface type and number
Router(config)#ip access-group 100 in
^
% Invalid input detected at '^' marker.

Router(config)#
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#access-list 100 deny ip 192.168.40.0 0.0.0.255 192.168.10.0 0.0.0.255
Router(config)#access-list 100 permit ip any any
Router(config)#interface g0/0.40
%Invalid interface type and number
Router(config)#interface g0/0/0.40
Router(config-subif)#ip access-group 100 in
Router(config-subif)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show access-lists
Extended IP access list 100
 10 deny ip 192.168.40.0 0.0.0.255 192.168.10.0 0.0.0.255
 20 permit ip any any
Router#
```

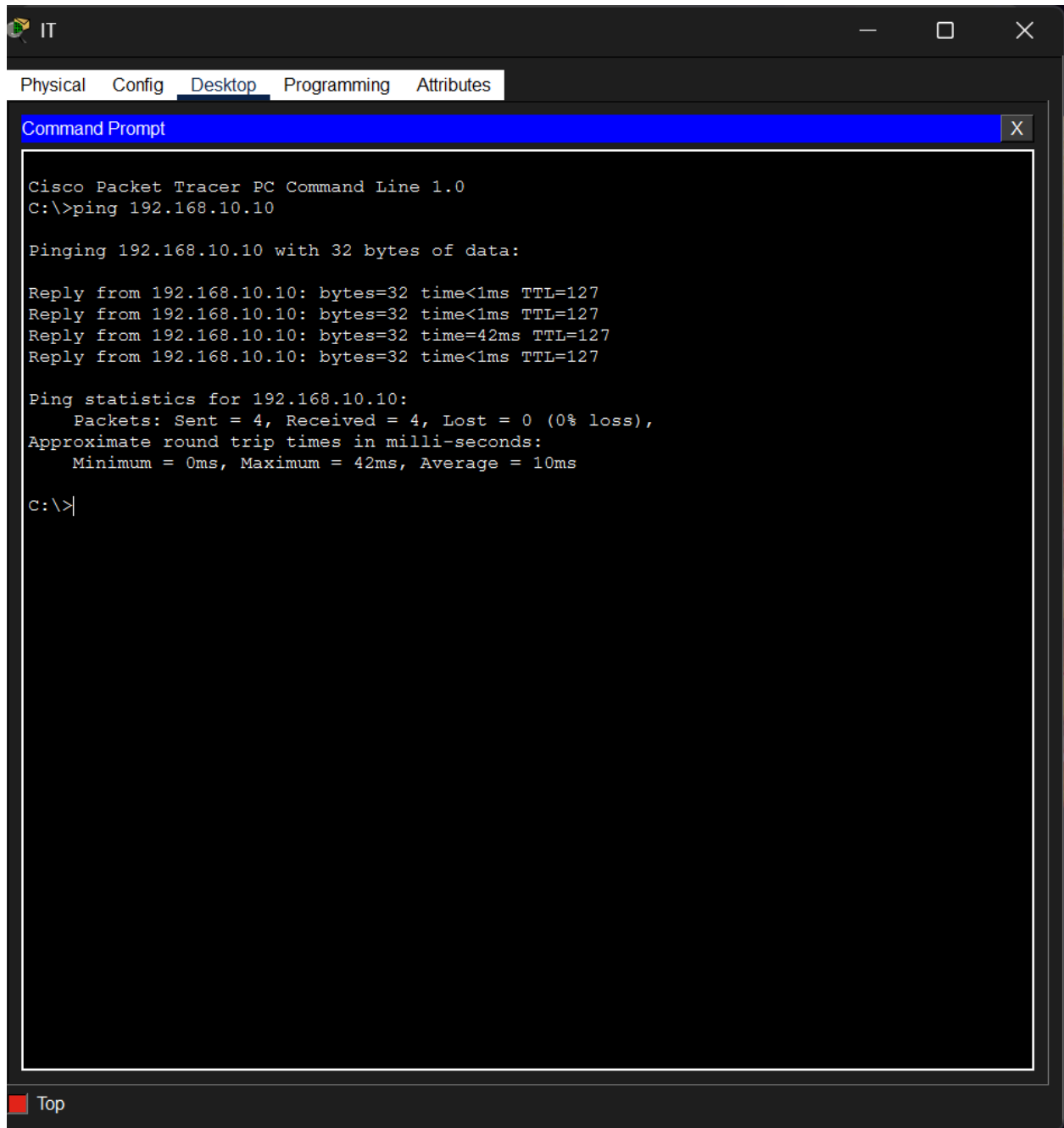
At the bottom of the window, there are "Copy" and "Paste" buttons, and a "Top" button with a red square icon.

ACL Configuration (show access-lists)

Security Testing (After ACL)

Testing confirmed that:

- Guest → Admin communication was blocked
- Other VLAN communication remained functional



The image shows a Cisco Packet Tracer PC Command Line window. The window has a title bar with 'IT' and standard window controls. Below the title bar are tabs for 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes'. The 'Desktop' tab is active, and within it, a 'Command Prompt' window is open. The command prompt shows the execution of a ping command to 192.168.10.10. The output indicates that the ping was successful, with 4 packets sent and received, 0% loss, and an average round trip time of 10ms. The window also features a 'Top' button at the bottom left.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

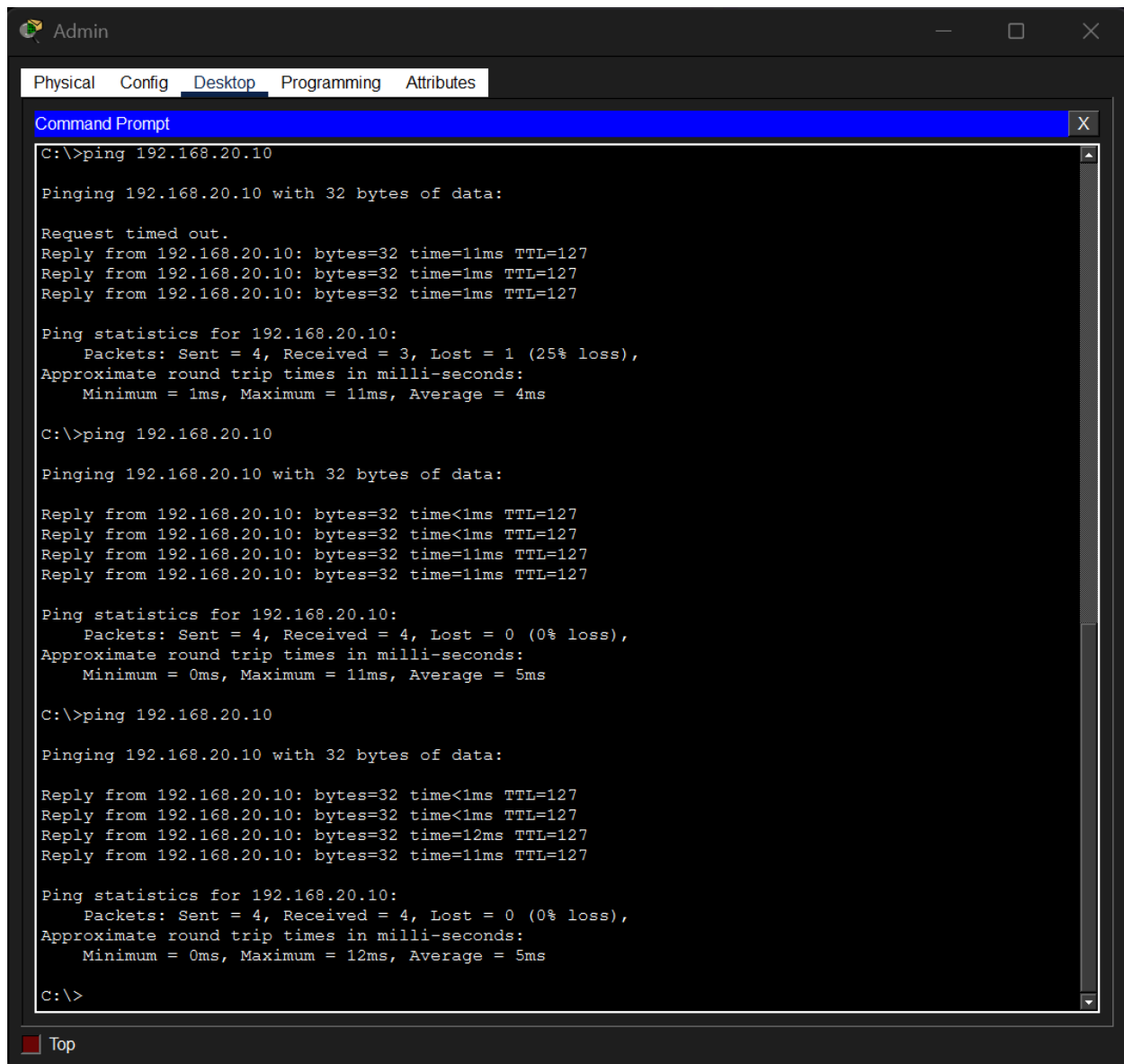
Reply from 192.168.10.10: bytes=32 time<1ms TTL=127
Reply from 192.168.10.10: bytes=32 time<1ms TTL=127
Reply from 192.168.10.10: bytes=32 time=42ms TTL=127
Reply from 192.168.10.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 42ms, Average = 10ms

C:\>
```

Top

Ping Test After ACL (Access Denied)



```
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.10: bytes=32 time=11ms TTL=127
Reply from 192.168.20.10: bytes=32 time=1ms TTL=127
Reply from 192.168.20.10: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 11ms, Average = 4ms

C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time=11ms TTL=127
Reply from 192.168.20.10: bytes=32 time=11ms TTL=127

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 5ms

C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time=12ms TTL=127
Reply from 192.168.20.10: bytes=32 time=11ms TTL=127

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 12ms, Average = 5ms

C:\>
```

Top

Allowed Communication After ACL

Results & Analysis

Testing Results

- Inter-VLAN communication worked successfully before applying ACL
- ACL correctly blocked unauthorized traffic
- Allowed traffic remained unaffected

Problem Resolution

Initial configuration issues were encountered with router interfaces due to incorrect interface naming. This was resolved by using the correct interface (g0/0/0) for subinterfaces.

Analysis

The implementation significantly improved network security by:

- Segmenting the network
- Reducing unauthorized access
- Applying controlled communication policies

The solution demonstrated how simple configurations can greatly enhance cybersecurity posture.

Conclusion & Recommendations

Summary of Outcomes

The project successfully achieved its objectives by designing and implementing a secure network with VLAN segmentation and access control.

Lessons Learned

- Importance of correct interface configuration
- Role of VLANs in network security
- How ACLs enforce access control
- Value of testing before and after implementation

Recommendations

- Implement firewall solutions
- Add intrusion detection systems (IDS/IPS)
- Enable multi-factor authentication (MFA)
- Perform regular vulnerability scans
- Monitor logs using SIEM tools

References

- Cisco Networking Documentation
- Cybersecurity Bootcamp Materials
- NIST Cybersecurity Framework
- MITRE ATT&CK Framework